CLASS 10 — MATHS · FORMULA SHEET

Chapter 4: **Quadratic Equations** | Quick revision, one page

Standard form

- $ax^2 + bx + c = 0$, where a, b, c are real and $a \neq 0$. A root α satisfies $a\alpha^2 + b\alpha + c = 0$.
- A quadratic equation has at most 2 roots.

Solving by factorisation

- Split the middle term: find two numbers with product ac and sum b; factorise; put each factor equal to zero.
- Example: $x^2 - 5x + 6 = (x-2)(x-3) = 0 \Rightarrow x = 2, 3$.

Quadratic formula

- $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, valid when $b^2 - 4ac \geq 0$.

Discriminant $D = b^2 - 4ac$

- $D > 0$: two distinct real roots.
- $D = 0$: two equal real roots, $x = -\dfrac{b}{2a}$.
- $D < 0$: no real roots.

Word problems

- Form the equation from the story (area, consecutive integers, speed-distance-time, age, etc.).
- Time $= \dfrac{\text{distance}}{\text{speed}}$; area of rectangle $= l \times b$.
- Reject roots that are negative or otherwise impossible for the situation, and verify the answer.

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